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Master of Computer Applications (MCA)

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Functional & Non-Functional Requirements, Use Cases & Activity Diagrams

HireHub: Premium Job Portal

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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

1. Introduction

1.1 Purpose

This document describes the Software Requirements Specification for HireHub: Premium Job Portal, a full-stack web application designed to connect job seekers with employers through an intelligent and modern recruitment platform.

1.2 Scope

The system serves three user types: Job Seekers (search, apply, track), Employers (post, manage, assess), and Administrators (manage, monitor). The platform provides REST APIs consumed by the React frontend, with real-time features via WebSocket.

2. Functional Requirements

2.1 User Authentication (FR-1)

ID	Requirement	Priority
FR-1.1	Users can register with email and password	Must
FR-1.2	Users can login with email/password	Must
FR-1.3	Users can login with Google OAuth2	Should
FR-1.4	JWT tokens expire after 24 hours	Must
FR-1.5	Users can logout securely, clearing the session	Must
FR-1.6	Password reset via email link	Should

2.2 User Profile Management (FR-2)

ID	Requirement	Priority
FR-2.1	Job seekers can create and edit personal profile	Must
FR-2.2	Employers can create and edit company profile	Must
FR-2.3	Users can upload a profile picture	Should
FR-2.4	Users can view other public user profiles	Should

2.3 Job Management (FR-3)

ID	Requirement	Priority
FR-3.1	Employers can post new jobs with full details	Must
FR-3.2	Employers can edit and delete their job postings	Must
FR-3.3	Job seekers can search jobs with multiple filters	Must
FR-3.4	Job seekers can save jobs for later review	Should
FR-3.5	Jobs support filters: location, salary, type, experience	Must

2.4 Resume Management (FR-4)

ID	Requirement	Priority
FR-4.1	Users can build resume using guided step-by-step builder	Must
FR-4.2	Users can upload PDF resumes from their device	Must
FR-4.3	System should analyse ATS compatibility and return score	Should
FR-4.4	Users can manage multiple resumes and set a primary one	Must

2.5 Application Management (FR-5)

ID	Requirement	Priority
FR-5.1	Job seekers can apply to jobs with a selected resume	Must
FR-5.2	Job seekers can track their application status pipeline	Must
FR-5.3	Employers can view all received applications per job	Must
FR-5.4	Employers can update application status through pipeline	Must
FR-5.5	System calculates AI match score for each application	Should
FR-5.6	Kanban view for employer application pipeline management	Should

2.6 Messaging System (FR-6)

ID	Requirement	Priority
FR-6.1	Users can send text messages to connected users	Must
FR-6.2	Real-time message delivery via WebSocket (STOMP)	Must
FR-6.3	Complete message history is preserved in database	Must
FR-6.4	Users can only initiate messaging after an application exists	Must

2.7 Notification System (FR-7)

ID	Requirement	Priority
FR-7.1	Users receive notifications for new messages	Must
FR-7.2	Users receive notifications for application status changes	Must
FR-7.3	Notifications are delivered in real-time via WebSocket	Must
FR-7.4	Users can mark individual or all notifications as read	Must

2.8 Quiz System (FR-8)

ID	Requirement	Priority
FR-8.1	Employers can create quizzes linked to job postings	Should
FR-8.2	Employers can add multiple-choice questions to quizzes	Should
FR-8.3	Job seekers can take assigned quizzes with time limits	Must
FR-8.4	System calculates quiz scores automatically on submission	Must
FR-8.5	Quiz results are visible to employers in the application view	Must

2.9 Interview Scheduling (FR-9)

ID	Requirement	Priority
FR-9.1	Employers can schedule interviews for shortlisted candidates	Should
FR-9.2	Job seekers can view their scheduled interview details	Must
FR-9.3	Employers can mark interviews as complete with feedback	Should

2.10 Analytics & Dashboard (FR-10)

ID	Requirement	Priority
FR-10.1	Dashboard shows key statistics (active jobs, applications, etc.)	Must
FR-10.2	Charts display application trends and hiring pipeline data	Should
FR-10.3	Employers can view company-specific job and applicant analytics	Should

3. Non-Functional Requirements

3.1 Performance

ID	Requirement	Metric
NFR-1.1	Page load time	< 3 seconds
NFR-1.2	API response time	< 500 ms
NFR-1.3	Support concurrent users	100+
NFR-1.4	Database query time	< 200 ms

3.2 Scalability

ID	Requirement
NFR-2.1	System should handle 10,000+ job postings without degradation
NFR-2.2	System should support 1,000+ concurrent users at peak load

3.3 Security

ID	Requirement
NFR-3.1	All passwords must be hashed using BCrypt before storage
NFR-3.2	JWT tokens must be signed with HMAC-SHA256 secret
NFR-3.3	All client-server communication must use HTTPS
NFR-3.4	Role-based access control enforced at every API endpoint
NFR-3.5	All user inputs must be validated and sanitised server-side

3.4 Reliability

ID	Requirement
NFR-4.1	System uptime target: 99% or above
NFR-4.2	Automated daily database backup
NFR-4.3	Graceful error handling with meaningful error messages

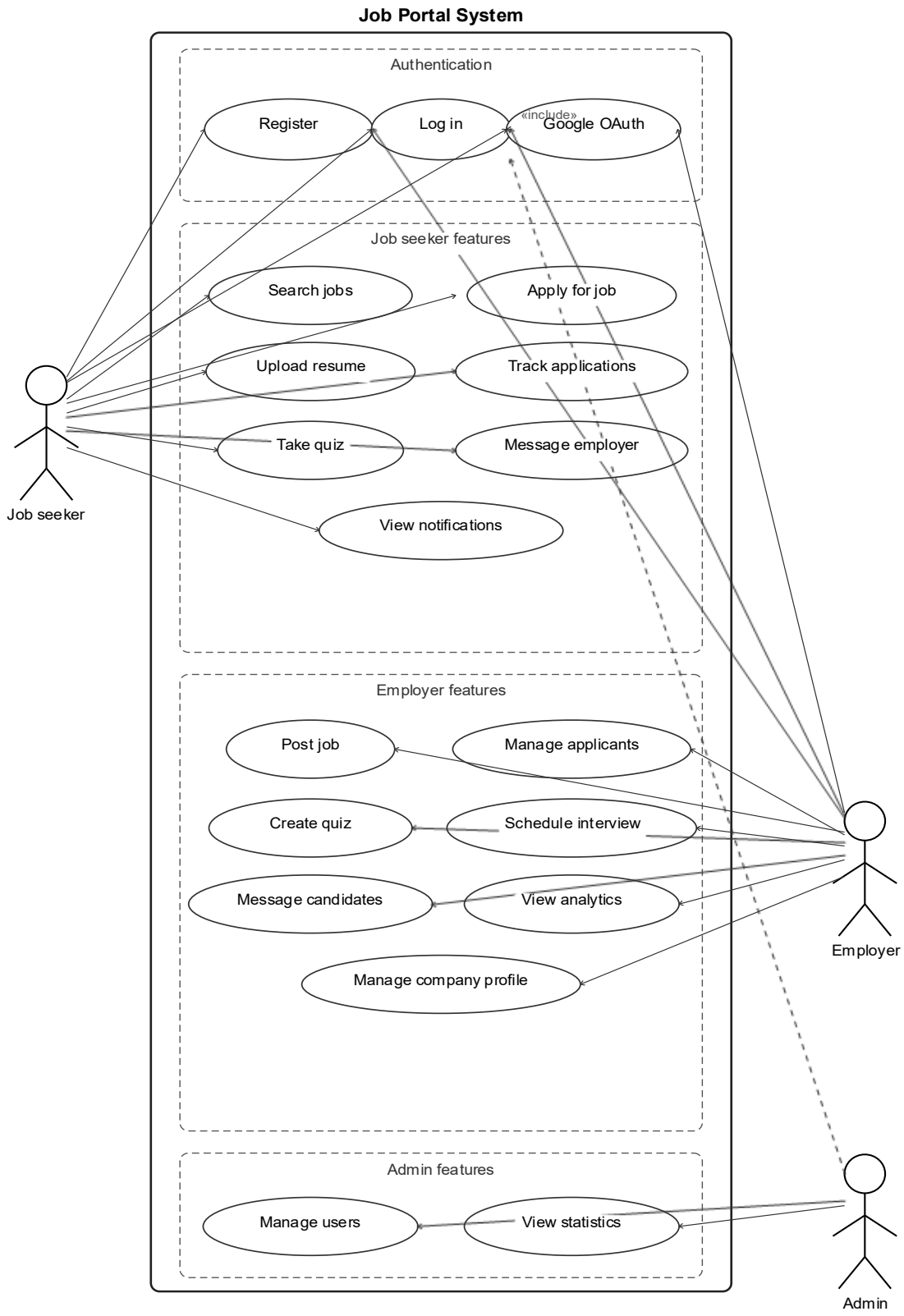
3.5 Usability

ID	Requirement
NFR-5.1	Responsive design that works on mobile, tablet, and desktop
NFR-5.2	Dark mode and light mode support with user preference persistence
NFR-5.3	Intuitive navigation with maximum 3-click access to core features
NFR-5.4	Accessible colour contrast ratio (WCAG 2.1 AA standard)

3.6 Browser Compatibility

ID	Browser
NFR-6.1	Google Chrome (latest stable version)
NFR-6.2	Mozilla Firefox (latest stable version)
NFR-6.3	Microsoft Edge (latest stable version)

Use Case Diagram



5. Use Case Descriptions

UC-1: User Registration

Actors: Job Seeker, Employer

Flow: (1) User clicks Register. (2) User selects account type (Job Seeker / Employer). (3) User enters email, password, and name. (4) System validates input fields. (5) System creates the user account in the database. (6) System sends a verification email. (7) User verifies email. (8) Account is activated and ready to use.

UC-2: Job Search and Application

Actors: Job Seeker

Flow: (1) User enters search criteria. (2) System displays matching job listings. (3) User selects a job and views details. (4) User selects a resume from their profile. (5) User submits the application. (6) System calculates the AI match score. (7) System sends a notification to the employer.

UC-3: Resume Building

Actors: Job Seeker

Flow: (1) User opens the Resume Builder. (2) User enters personal details. (3) User adds education entries. (4) User adds work experience. (5) User adds skills. (6) User adds projects and certifications. (7) System generates a downloadable PDF. (8) User saves the resume to their profile.

UC-4: Application Management

Actors: Employer

Flow: (1) Employer views received applications. (2) Employer filters by status or job. (3) Employer reviews candidate profile and resume. (4) Employer updates application status. (5) Employer schedules an interview. (6) System notifies the candidate of the status change.

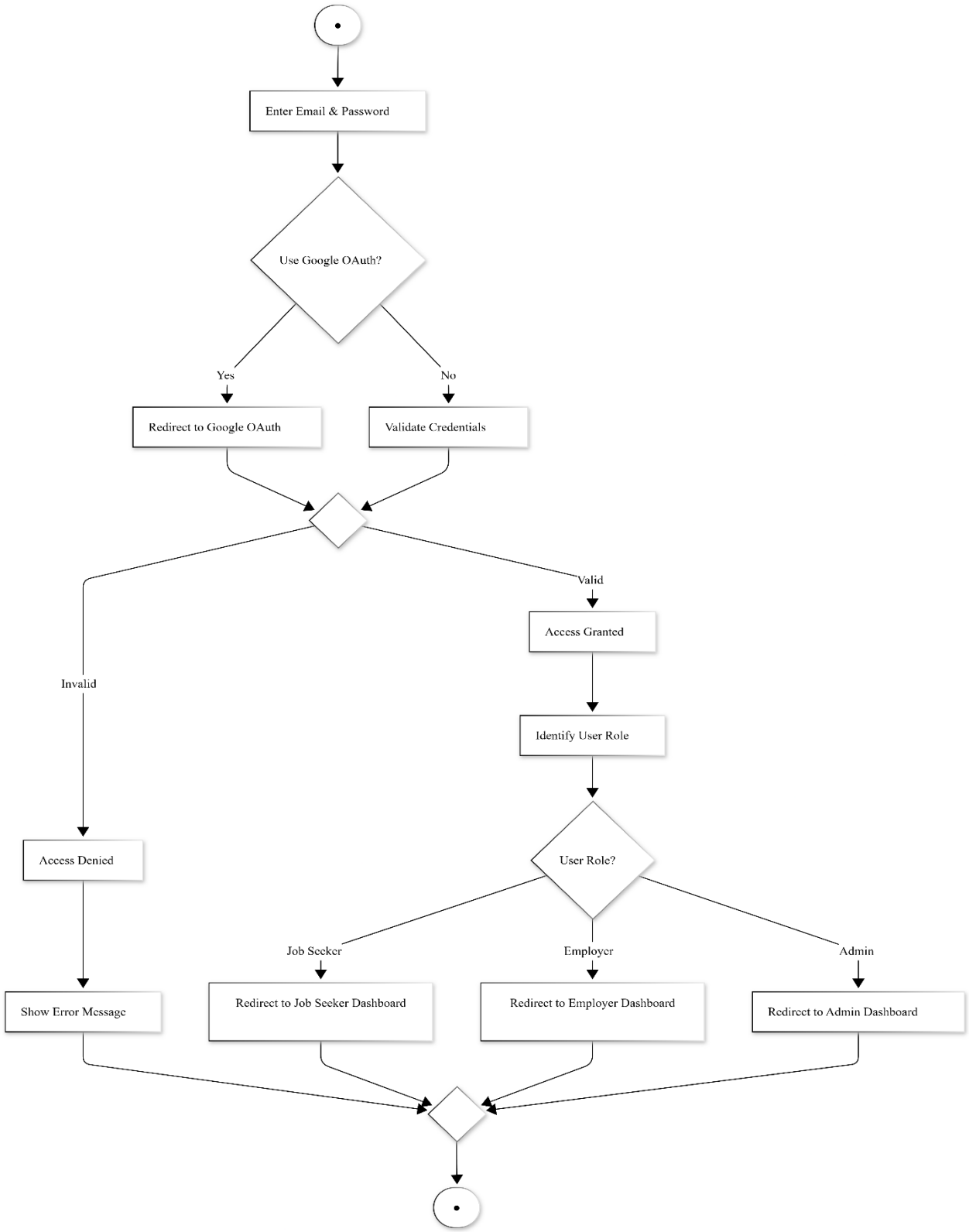
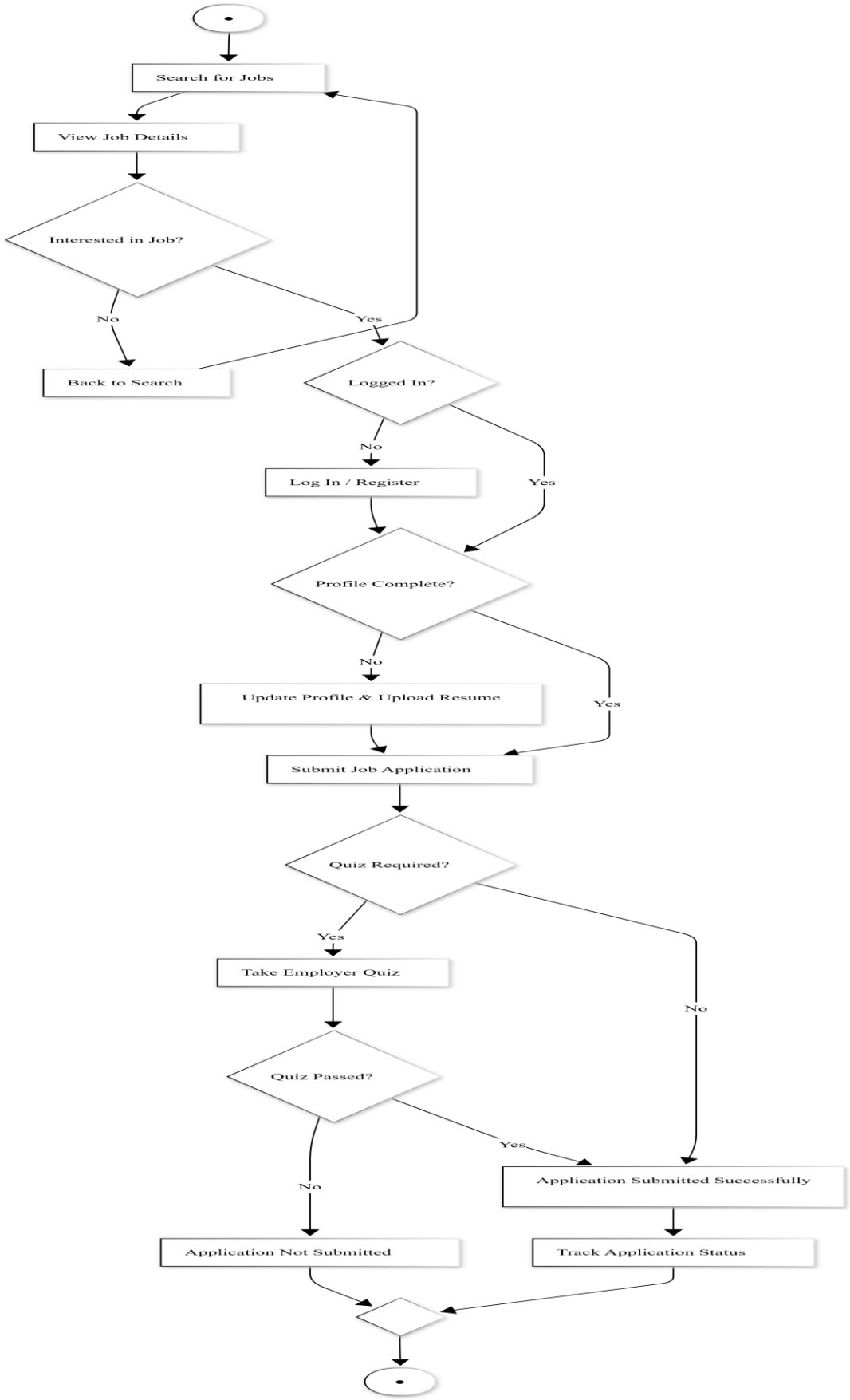
UC-5: Real-time Messaging

Actors: Job Seeker, Employer

Flow: (1) User opens the chat interface. (2) System establishes a WebSocket (STOMP) connection. (3) User types and sends a message. (4) System delivers the message in real-time via the message broker. (5) Recipient receives the message instantly. (6) Message is saved to the database for history.

Activity Diagrams

Login Activity Diagram



Job Application Activity Diagram